

- 2 A uniform rectangular card is suspended from a wooden rod. The card is held at one of its ends as shown in Fig. 2.1. The force by the hand on the card acts horizontally to the right.

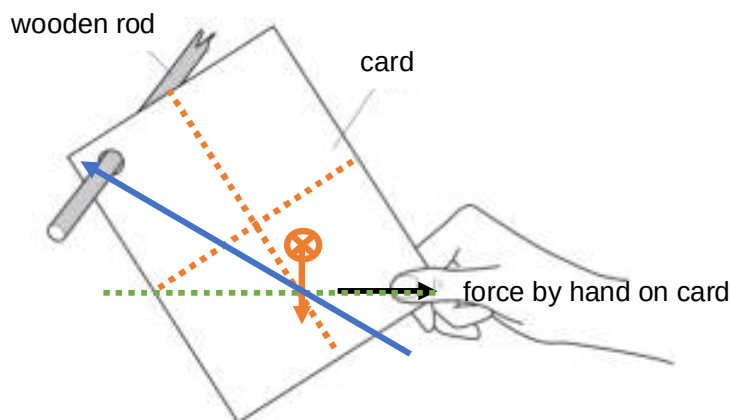


Fig. 2.1

(a) On Fig. 2.1,

- (i) mark with an 'X' the position of the centre of gravity of the card. [1]

Accept answers that off-centre (below and towards the right) [1]

- (ii) draw an arrow labelled with W to represent the weight of the card. [1]

arrow points downwards starting from X [1]

(b) State the conditions for the card to be in equilibrium.

... Net force is zero [B1]
 ... Net torque is zero [B1] or the sum of clockwise moments about any point is equal
 ... to the sum of anticlockwise moments about that same point or the lines of action
 of the 3 forces (weight, force by hand on card and force by rod on card) passes
 through a common point [2]

(c) Draw an arrow labelled with R on Fig. 2.1 to represent the force exerted by the wooden rod on the card.

Show your construction clearly. [2]

Concurrent forces, 3 forces passes through same point. [B1]
 Correct direction of arrow [B1]

(d) The card is now released. It swings on the wooden rod and eventually comes to a rest.

By reference to the completed diagram in Fig. 2.1, describe the final position in which the card comes to a rest.

Position of X directly below point of contact of card with rod/ line of action of W
 passing through the point of contact of card with rod [1]
 Such that there is no resultant moment
 Correct diagrams are accepted as part of the working. [1]

2 (a) An object is placed on a smooth horizontal surface and is connected to a light spring, as